

**INTERNATIONAL CYBERSECURITY  
AND DIGITAL FORENSICS ACADEMY**

**INTRO TO CRITICAL INFRASTRUCTURE  
FOR IT LEARNERS**

**LAB 07 — EXPLORING ICS PACKET CAPTURES**

**Student Edition**

Authorized ICDFA training environments only  
[icdfa.edu.ng](http://icdfa.edu.ng)

## Lab 07: Exploring ICS Packet Captures

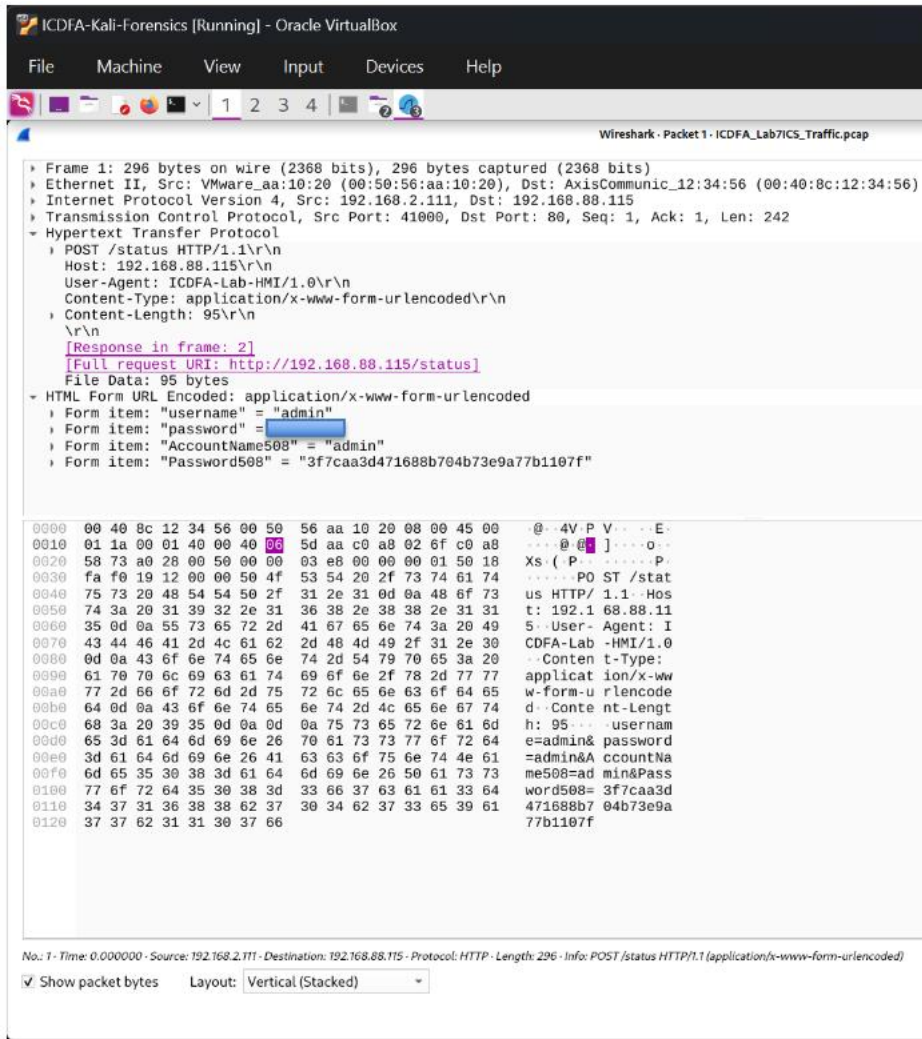
**Course:** Intro to Critical Infrastructure for IT Learners | **Estimated Duration:** 4–5 hours

**Submission deadline:** As published in ICDFA E-Campus.

**Authorization & Safety:** Perform all activities only on the ICDFA-provided virtual machines, packet captures and isolated training networks. Do not bridge these VMs to production networks, scan external systems, or reuse credentials observed in training data.

### Required Evidence & Submission

- At least six protocol-analysis screenshots, one for each required protocol family.



### Screenshot 1 (HTTP) with Redaction

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)  
INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

ICDFA-Kali-Forensics [Running] - Oracle VirtualBox

File Machine View Input Devices Help

ICDFA\_Lab7ICS\_Traffic.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Current filter: ftp

| No.   | Time     | Source        | Destination   | Protocol | Length | Info                                                        |
|-------|----------|---------------|---------------|----------|--------|-------------------------------------------------------------|
| 12004 | 9.002249 | 192.168.2.206 | 10.10.10.50   | FTP      | 77     | Response: 331 Password required                             |
| 12005 | 9.003000 | 10.10.10.50   | 192.168.2.206 | FTP      | 60     | Request: SYST                                               |
| 12006 | 9.003750 | 192.168.2.206 | 10.10.10.50   | FTP      | 74     | Response: 230 User logged in                                |
| 12007 | 9.004499 | 10.10.10.50   | 192.168.2.206 | FTP      | 59     | Request: PWD                                                |
| 12008 | 9.005249 | 192.168.2.206 | 10.10.10.50   | FTP      | 73     | Response: 215 UNIX Type: L8                                 |
| 12009 | 9.006000 | 10.10.10.50   | 192.168.2.206 | FTP      | 60     | Request: LIST                                               |
| 12010 | 9.006750 | 192.168.2.206 | 10.10.10.50   | FTP      | 92     | Response: 257 "/usr/html" is current directory              |
| 12011 | 9.007499 | 10.10.10.50   | 192.168.2.206 | FTP      | 72     | Request: USER cameraadmin                                   |
| 12012 | 9.008249 | 192.168.2.206 | 10.10.10.50   | FTP      | 165    | Response: 220 AXIS 206 Network Camera 4.40 FTP Server ready |
| 12013 | 9.009000 | 10.10.10.50   | 192.168.2.206 | FTP      | 74     | Request: PASS axis-lab-2026                                 |
| 12014 | 9.009749 | 192.168.2.206 | 10.10.10.50   | FTP      | 77     | Response: 331 Password required                             |
| 12015 | 9.010499 | 10.10.10.50   | 192.168.2.206 | FTP      | 60     | Request: SYST                                               |
| 12016 | 9.011250 | 192.168.2.206 | 10.10.10.50   | FTP      | 74     | Response: 230 User logged in                                |
| 12017 | 9.012000 | 10.10.10.50   | 192.168.2.206 | FTP      | 59     | Request: PWD                                                |
| 12018 | 9.012749 | 192.168.2.206 | 10.10.10.50   | FTP      | 73     | Response: 215 UNIX Type: L8                                 |
| 12019 | 9.013499 | 10.10.10.50   | 192.168.2.206 | FTP      | 60     | Request: LIST                                               |
| 12020 | 9.014250 | 192.168.2.206 | 10.10.10.50   | FTP      | 92     | Response: 257 "/usr/html" is current directory              |
| 12021 | 9.015000 | 10.10.10.50   | 192.168.2.206 | FTP      | 72     | Request: USER cameraadmin                                   |
| 12022 | 9.015749 | 192.168.2.206 | 10.10.10.50   | FTP      | 165    | Response: 220 AXIS 206 Network Camera 4.40 FTP Server ready |
| 12023 | 9.016499 | 10.10.10.50   | 192.168.2.206 | FTP      | 74     | Request: PASS                                               |
| 12024 | 9.017250 | 192.168.2.206 | 10.10.10.50   | FTP      | 77     | Response: 331 Password required                             |
| 12025 | 9.017999 | 10.10.10.50   | 192.168.2.206 | FTP      | 60     | Request: SYST                                               |

Frame 12021: 72 bytes on wire (576 bits), 72 bytes captured (576 bits)

Ethernet II, Src: PCSSystemtec\_11:22:33 (08:00:27:11:22:33), Dst: AxisCommunic\_08:00:27:11:22:33

Internet Protocol Version 4, Src: 10.10.10.50, Dst: 192.168.2.206

Transmission Control Protocol, Src Port: 42916, Dst Port: 21, Seq: 1, Ack: 1, Len: 72

File Transfer Protocol (FTP)

USER cameraadmin\r\n

Request command: USER

Request arg: cameraadmin

[Current working directory: ]

Current working directory (ftp.current-working-directory)

Packets: 302000 · Displayed: 3000 (1.0%) Profile: Default

Screenshot 2 (FTP)

**INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)**  
INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

ICDFA-Kali-Forensics [Running] - Oracle VirtualBox

File Machine View Input Devices Help

ICDFA\_Lab7ICS\_Traffic.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

modbus

| No.   | Time      | Source        | Destination | Protocol  | Length | Info                                                               |
|-------|-----------|---------------|-------------|-----------|--------|--------------------------------------------------------------------|
| 15001 | 11.250000 | 192.168.2.111 | 10.10.20.12 | Modbus... | 65     | Query: Trans: 0; Unit: 1, Func: 43/ 1: Read Device Identification  |
| 15002 | 11.250750 | 192.168.2.111 | 10.10.20.12 | Modbus... | 66     | Query: Trans: 1; Unit: 1, Func: 1: Read Coils                      |
| 15003 | 11.251499 | 192.168.2.111 | 10.10.20.12 | Modbus... | 66     | Query: Trans: 2; Unit: 1, Func: 3: Read Holding Registers          |
| 15004 | 11.252249 | 192.168.2.111 | 10.10.20.12 | Modbus... | 66     | Query: Trans: 3; Unit: 1, Func: 6: Write Single Register           |
| 15005 | 11.253000 | 192.168.2.111 | 10.10.20.12 | Modbus... | 65     | Query: Trans: 4; Unit: 1, Func: 43/ 1: Read Device Identification  |
| 15006 | 11.253750 | 192.168.2.111 | 10.10.20.12 | Modbus... | 66     | Query: Trans: 5; Unit: 1, Func: 1: Read Coils                      |
| 15007 | 11.254499 | 192.168.2.111 | 10.10.20.12 | Modbus... | 66     | Query: Trans: 6; Unit: 1, Func: 3: Read Holding Registers          |
| 15008 | 11.255249 | 192.168.2.111 | 10.10.20.12 | Modbus... | 66     | Query: Trans: 7; Unit: 1, Func: 6: Write Single Register           |
| 15009 | 11.256000 | 192.168.2.111 | 10.10.20.12 | Modbus... | 65     | Query: Trans: 8; Unit: 1, Func: 43/ 1: Read Device Identification  |
| 15010 | 11.256750 | 192.168.2.111 | 10.10.20.12 | Modbus... | 66     | Query: Trans: 9; Unit: 1, Func: 1: Read Coils                      |
| 15011 | 11.257499 | 192.168.2.111 | 10.10.20.12 | Modbus... | 66     | Query: Trans: 10; Unit: 1, Func: 3: Read Holding Registers         |
| 15012 | 11.258249 | 192.168.2.111 | 10.10.20.12 | Modbus... | 66     | Query: Trans: 11; Unit: 1, Func: 6: Write Single Register          |
| 15013 | 11.259000 | 192.168.2.111 | 10.10.20.12 | Modbus... | 65     | Query: Trans: 12; Unit: 1, Func: 43/ 1: Read Device Identification |
| 15014 | 11.259749 | 192.168.2.111 | 10.10.20.12 | Modbus... | 66     | Query: Trans: 13; Unit: 1, Func: 1: Read Coils                     |
| 15015 | 11.260499 | 192.168.2.111 | 10.10.20.12 | Modbus... | 66     | Query: Trans: 14; Unit: 1, Func: 3: Read Holding Registers         |
| 15016 | 11.261250 | 192.168.2.111 | 10.10.20.12 | Modbus... | 66     | Query: Trans: 15; Unit: 1, Func: 6: Write Single Register          |
| 15017 | 11.262000 | 192.168.2.111 | 10.10.20.12 | Modbus... | 65     | Query: Trans: 16; Unit: 1, Func: 43/ 1: Read Device Identification |
| 15018 | 11.262749 | 192.168.2.111 | 10.10.20.12 | Modbus... | 66     | Query: Trans: 17; Unit: 1, Func: 1: Read Coils                     |
| 15019 | 11.263499 | 192.168.2.111 | 10.10.20.12 | Modbus... | 66     | Query: Trans: 18; Unit: 1, Func: 3: Read Holding Registers         |
| 15020 | 11.264250 | 192.168.2.111 | 10.10.20.12 | Modbus... | 66     | Query: Trans: 19; Unit: 1, Func: 6: Write Single Register          |
| 15021 | 11.265000 | 192.168.2.111 | 10.10.20.12 | Modbus... | 65     | Query: Trans: 20; Unit: 1, Func: 43/ 1: Read Device Identification |
| 15022 | 11.265749 | 192.168.2.111 | 10.10.20.12 | Modbus... | 66     | Query: Trans: 21; Unit: 1, Func: 1: Read Coils                     |

Frame 15001: 65 bytes on wire (520 bits), 65 bytes captured (520 bits) on 0  
Ethernet II, Src: VMware\_aa:10:20 (00:50:56:aa:10:20), Dst: Siemens\_44:55:66 (00:0e:0c:00:00:00)  
Internet Protocol Version 4, Src: 192.168.2.111, Dst: 10.10.20.12  
Transmission Control Protocol, Src Port: 45000, Dst Port: 502, Seq: 1, Ack: 1, Len: 1  
Modbus/TCP  
Modbus  
010 1011 = Function Code: Encapsulated Interface Transport (43)  
MEI type: Read Device Identification (14)  
Read Device ID: Basic Device Identification (1)  
Object ID: VendorName (0)

Packets: 302000 - Displayed: 100000 (33.1%) Profile: Default

**Screenshot 3 (Modbus)**

**INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)**  
INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

ICDFA-Kali-Forensics [Running] - Oracle VirtualBox

File Machine View Input Devices Help

ICDFA\_Lab7ICS\_Traffic.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.port == 20000 || udp.port == 20000

| No.   | Time      | Source      | Destination | Protocol | Length | Info                                                  |
|-------|-----------|-------------|-------------|----------|--------|-------------------------------------------------------|
| 11500 | 86.250000 | 10.10.10.50 | 10.10.30.20 | TCP      | 76     | 46000 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=64240 Len=22 |
| 11500 | 86.250750 | 10.10.10.50 | 10.10.30.20 | TCP      | 76     | 46001 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=64240 Len=22 |
| 11500 | 86.251499 | 10.10.10.50 | 10.10.30.20 | TCP      | 76     | 46002 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=64240 Len=22 |
| 11500 | 86.252249 | 10.10.10.50 | 10.10.30.20 | TCP      | 76     | 46003 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=64240 Len=22 |
| 11500 | 86.253000 | 10.10.10.50 | 10.10.30.20 | TCP      | 76     | 46004 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=64240 Len=22 |
| 11500 | 86.253750 | 10.10.10.50 | 10.10.30.20 | TCP      | 76     | 46005 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=64240 Len=22 |
| 11500 | 86.254499 | 10.10.10.50 | 10.10.30.20 | TCP      | 76     | 46006 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=64240 Len=22 |
| 11500 | 86.255249 | 10.10.10.50 | 10.10.30.20 | TCP      | 76     | 46007 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=64240 Len=22 |
| 11500 | 86.256000 | 10.10.10.50 | 10.10.30.20 | TCP      | 76     | 46008 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=64240 Len=22 |
| 11500 | 86.256750 | 10.10.10.50 | 10.10.30.20 | TCP      | 76     | 46009 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=64240 Len=22 |
| 11500 | 86.257499 | 10.10.10.50 | 10.10.30.20 | TCP      | 76     | 46010 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=64240 Len=22 |
| 11500 | 86.258249 | 10.10.10.50 | 10.10.30.20 | TCP      | 76     | 46011 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=64240 Len=22 |
| 11500 | 86.259000 | 10.10.10.50 | 10.10.30.20 | TCP      | 76     | 46012 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=64240 Len=22 |
| 11500 | 86.259749 | 10.10.10.50 | 10.10.30.20 | TCP      | 76     | 46013 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=64240 Len=22 |
| 11500 | 86.260499 | 10.10.10.50 | 10.10.30.20 | TCP      | 76     | 46014 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=64240 Len=22 |
| 11500 | 86.261250 | 10.10.10.50 | 10.10.30.20 | TCP      | 76     | 46015 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=64240 Len=22 |
| 11500 | 86.262000 | 10.10.10.50 | 10.10.30.20 | TCP      | 76     | 46016 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=64240 Len=22 |
| 11500 | 86.262749 | 10.10.10.50 | 10.10.30.20 | TCP      | 76     | 46017 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=64240 Len=22 |
| 11500 | 86.263499 | 10.10.10.50 | 10.10.30.20 | TCP      | 76     | 46018 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=64240 Len=22 |
| 11500 | 86.264250 | 10.10.10.50 | 10.10.30.20 | TCP      | 76     | 46019 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=64240 Len=22 |
| 11500 | 86.265000 | 10.10.10.50 | 10.10.30.20 | TCP      | 76     | 46020 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=64240 Len=22 |
| 11500 | 86.265749 | 10.10.10.50 | 10.10.30.20 | TCP      | 76     | 46021 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=64240 Len=22 |

Frame 115001: 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on interface 0  
Ethernet II, Src: PCSSystemtec\_11:22:33 (08:00:27:11:22:33), Dst: Telemechaniqu\_77:8  
Internet Protocol Version 4, Src: 10.10.10.50, Dst: 10.10.30.20  
Transmission Control Protocol, Src Port: 46000, Dst Port: 20000, Seq: 1, Ack: 1, Len: 22  
Data (22 bytes)

ICDFA\_Lab7ICS\_Traffic.pcap

Packets: 302000 · Displayed: 28000 (9.3%) Profile: Default

**Screenshot 4 (DNP3).**

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)  
INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

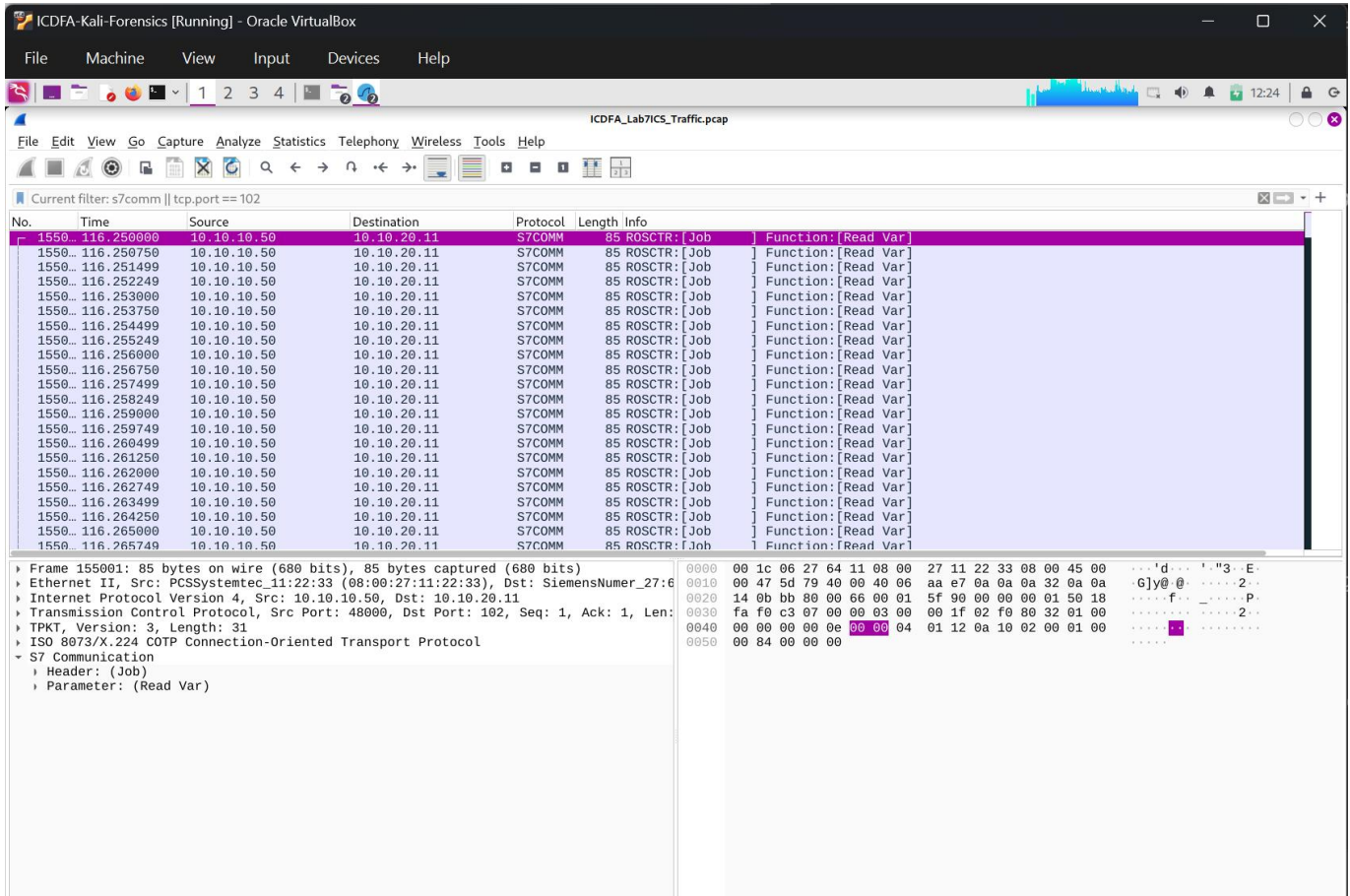
The screenshot displays the ICDFA-Kali-Forensics [Running] - Oracle VirtualBox window. The main application is Wireshark, which is capturing network traffic from the file 'ICDFA\_Lab7ICS\_Traffic.pcap'. The filter bar shows the active filter: 'cip || enip || tcp.port == 44818 || udp.port == 2222'. The packet list pane shows a series of packets, with packet 143001 selected. The packet details pane shows the structure of the selected packet, including Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and EtherNet/IP (Industrial Protocol). The packet bytes pane shows the raw data in hexadecimal and ASCII. The status bar at the bottom indicates that 302000 packets are displayed, with 12000 (4.0%) shown. The profile is set to Default.

| No.    | Time       | Source      | Destination | Protocol | Length | Info                                        |
|--------|------------|-------------|-------------|----------|--------|---------------------------------------------|
| 143001 | 107.250000 | 10.10.10.50 | 10.10.20.11 | ENIP     | 82     | Register Session (Req), Session: 0x11223344 |
| 143002 | 107.250750 | 10.10.10.50 | 10.10.20.11 | CIP      | 98     | [Malformed Packet]                          |
| 143003 | 107.251499 | 10.10.10.50 | 10.10.20.11 | CIP      | 98     | [Malformed Packet]                          |
| 143004 | 107.252249 | 10.10.10.50 | 10.10.20.11 | CIP      | 98     | [Malformed Packet]                          |
| 143005 | 107.253000 | 10.10.10.50 | 10.10.20.11 | CIP      | 98     | [Malformed Packet]                          |
| 143006 | 107.253750 | 10.10.10.50 | 10.10.20.11 | CIP      | 98     | [Malformed Packet]                          |
| 143007 | 107.254499 | 10.10.10.50 | 10.10.20.11 | CIP      | 98     | [Malformed Packet]                          |
| 143008 | 107.255249 | 10.10.10.50 | 10.10.20.11 | CIP      | 98     | [Malformed Packet]                          |
| 143009 | 107.256000 | 10.10.10.50 | 10.10.20.11 | CIP      | 98     | [Malformed Packet]                          |
| 143010 | 107.256750 | 10.10.10.50 | 10.10.20.11 | CIP      | 98     | [Malformed Packet]                          |
| 143011 | 107.257499 | 10.10.10.50 | 10.10.20.11 | CIP      | 98     | [Malformed Packet]                          |
| 143012 | 107.258249 | 10.10.10.50 | 10.10.20.11 | CIP      | 98     | [Malformed Packet]                          |
| 143013 | 107.259000 | 10.10.10.50 | 10.10.20.11 | CIP      | 98     | [Malformed Packet]                          |
| 143014 | 107.259749 | 10.10.10.50 | 10.10.20.11 | CIP      | 98     | [Malformed Packet]                          |
| 143015 | 107.260499 | 10.10.10.50 | 10.10.20.11 | CIP      | 98     | [Malformed Packet]                          |
| 143016 | 107.261250 | 10.10.10.50 | 10.10.20.11 | CIP      | 98     | [Malformed Packet]                          |
| 143017 | 107.262000 | 10.10.10.50 | 10.10.20.11 | CIP      | 98     | [Malformed Packet]                          |
| 143018 | 107.262749 | 10.10.10.50 | 10.10.20.11 | CIP      | 98     | [Malformed Packet]                          |
| 143019 | 107.263499 | 10.10.10.50 | 10.10.20.11 | CIP      | 98     | [Malformed Packet]                          |
| 143020 | 107.264250 | 10.10.10.50 | 10.10.20.11 | CIP      | 98     | [Malformed Packet]                          |
| 143021 | 107.265000 | 10.10.10.50 | 10.10.20.11 | ENIP     | 82     | Register Session (Req), Session: 0x11223344 |
| 143022 | 107.265749 | 10.10.10.50 | 10.10.20.11 | CIP      | 98     | [Malformed Packet]                          |

Frame 143001: 82 bytes on wire (656 bits), 82 bytes captured (656 bits) on interface 0  
Ethernet II, Src: PCSSystemtec\_11:22:33 (08:00:27:11:22:33), Dst: SiemensNumer\_27:00:00:00:00:00  
Internet Protocol Version 4, Src: 10.10.10.50, Dst: 10.10.20.11  
Transmission Control Protocol, Src Port: 47000, Dst Port: 44818, Seq: 1, Ack: 1, Len: 82  
EtherNet/IP (Industrial Protocol), Session: 0x11223344, Register Session  
Encapsulation Header  
Command Specific Data

0000 00 1c 06 27 64 11 08 00 27 11 22 33 08 00 45 00 ...d...".3..E..  
0010 00 44 2e 99 40 00 40 06 d9 ca 0a 0a 0a 32 0a 0a ...D... ..2..  
0020 14 0b b7 98 af 12 00 01 11 70 00 00 00 01 50 18 ... ..P..  
0030 fa f0 3b 1c 00 00 65 00 04 00 44 33 22 11 00 00 ...e...D3"..  
0040 00 00 49 43 44 46 41 30 30 37 00 00 00 01 00 ...ICDFA0 07.....  
0050 00 00 .....

Screenshot 5 (Ethernet/IP & CIP)



**Screenshot 6 (s7comm)**

- **A Protocol Evidence Matrix with filters, endpoints, observed operations and security significance.**

| Protocol | Wireshark Display Filter                        | Observed Endpoints (Src to Destination) | Observed Operations/Functions                                                  | Security Significance                                                       |
|----------|-------------------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| HRRP     | http.authbasic or http.request.method == "POST" | 192.168.2.11 to 192.168.88.115          | POST /status HTTP/1.1 with cleartext username=admin and password=admin in body | Transmits credentials in cleartext; vulnerable to passive network sniffing. |
| FTP      | ftp                                             | 10.10.10.50                             | 220 AXIS 206                                                                   | Discloses vendor,                                                           |

|                    |             |                                                        |                                                                                                  |                                                                                                          |
|--------------------|-------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
|                    |             | to<br>192.168.2.2<br>06                                | Network<br>Camera 4.40<br>FTP Server<br>ready, USER<br>cameraadmin,<br>PASS<br>[REDACTED]        | model, and<br>firmware banners<br>(AXIS 206 v4.40);<br>passes cleartext<br>authentication.               |
| <b>Modbus</b>      | modbus      | SCADA<br>Workstation to PLC                            | Read Holding<br>Registers (FC3),<br>Read Input<br>Registers (FC4),<br>Write Single Coil<br>(FC5) | Total lack of<br>encryption and<br>authentication<br>allows arbitrary<br>command<br>injection.           |
| <b>DNP3</b>        | dnp3        | Master<br>Station to<br>Outstation<br>/ RTU            | Class 0/1/2/3<br>Data Polls,<br>Direct Operate<br>commands                                       | Telemetry and<br>control<br>commands sent<br>cleartext;<br>vulnerable to<br>spoofing without<br>DNP3 SA. |
| <b>EtherNet/IP</b> | enip or cip | Engineering<br>Workstation to Allen-<br>Bradley<br>PLC | Send RR Data,<br>Register Session,<br>CIP Read/Write<br>Tag Services                             | Exposes internal<br>PLC memory tags<br>and control state<br>over<br>unencrypted<br>Ethernet.             |
| <b>Siemens S7</b>  | s7comm      | Engineering<br>Station to<br>Siemens S7<br>PLC         | ROSCTR Job<br>(0x01), Read<br>Var (0x04),<br>Setup<br>Communication                              | Proprietary,<br>unencrypted<br>protocol<br>exposing PLC<br>memory blocks<br>(DBs) and system<br>states.  |

- **A one-page security assessment prioritizing the observed risks.**

### **Executive Summary & Risk Prioritization**

A passive packet analysis of the provided industrial control systems (ICS) capture dataset (ICDFA\_Lab7ICS\_Traffic.pcap) revealed significant systemic vulnerabilities across all monitored network segments. The primary risk observations, prioritized by severity, include:

1. **Cleartext & Obfuscated Credential Exposure (Critical):** HTTP POST requests (`http.request.method == "POST"`) transmit administrative credentials in unencrypted form (`username=admin, password=admin`). FTP sessions similarly expose usernames and passwords (`USER cameraadmin, PASS axis-lab-2026`) in plain text. Passive attackers can harvest these credentials without sending a single probe.
2. **Exposed Device Banners & Metadata (High):** FTP responses reveal exact device hardware, vendor, and firmware details (e.g., `AXIS 206 Network Camera 4.40`). Adversaries can leverage this metadata to locate targeted, public CVE exploits.
3. **Absence of Authentication in OT Fieldbus Protocols (High):** Modbus TCP, DNP3, EtherNet/IP (CIP), and Siemens S7comm operate entirely without session authentication or integrity verification. Any host on the subnet can inject arbitrary control commands (e.g., forcing digital output coils or modifying holding registers).
4. **Lack of Transport Confidentiality (Medium):** Process telemetry, analog sensor values, relay statuses, and engineering logic tags are transmitted cleartext. Passive observers can map physical operational topology and baseline normal operations for future disruption.
5. **Insufficient Purdue Model Segmentation (Medium):** The simultaneous presence of IT management traffic (HTTP, FTP) alongside OT control protocols (Modbus, DNP3, S7) on shared capture interfaces indicates improper network isolation.

## Knowledge Check

**1. Why does Base64 encoding not provide confidentiality for HTTP Basic authentication?**

Response: Base64 is an encoding algorithm, not an encryption method. It is designed purely for data formatting and can be reversed instantly by anyone capturing the traffic without needing a secret key.

**2. What information can device banners or protocol metadata reveal to an analyst?**

Response: Banners reveal vendor names, exact hardware model numbers, firmware versions, operating system details, and supported services, allowing analysts (or adversaries) to identify vulnerable legacy hardware.

**3. How do Modbus, DNP3, EtherNet/IP and S7 differ from ordinary user-facing protocols such as HTTP or FTP?**

Response: Industrial protocols are designed for real-time, deterministic machine-to-machine communication between controllers and field devices. Unlike IT protocols that increasingly incorporate session security (e.g., TLS/SSH), legacy ICS protocols prioritize low latency and reliability over native encryption and authentication.

**4. Why is passive packet analysis valuable for understanding an ICS environment?**

Response: Passive packet analysis allows security teams to map network assets, discover protocols, identify operational anomalies, and detect threat indicators without sending a single packet, eliminating the risk of crashing sensitive or legacy industrial devices.

**5. What compensating controls can reduce risk where legacy industrial protocols cannot be replaced immediately?**

Response: Compensating controls include strict network segmentation (Purdue Model architecture), firewalls with Industrial Deep Packet Inspection (DPI), OT-aware Intrusion Detection Systems (IDS), encrypted IPsec/VPN tunnels for inter-site transit, and jump boxes with multi-factor authentication (MFA).